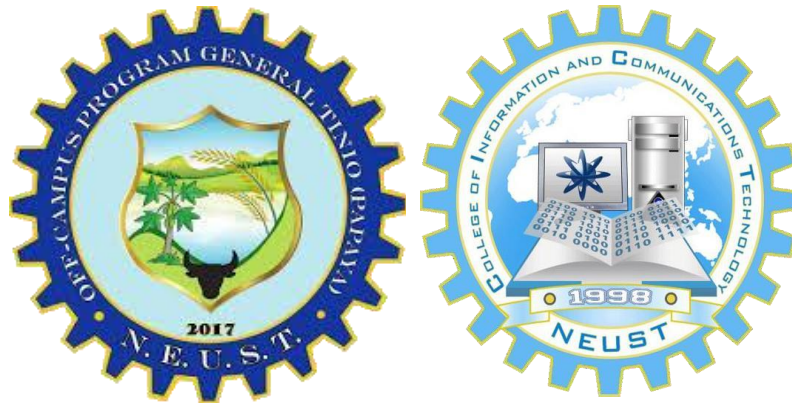


**"Design and Development of a Web-Based Integrated Enrollment and
Academic Management System for Palale High School**



A CAPSTONE AND RESEARCH PROJECT

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CHAPTER 1

Introduction

THE PROBLEM AND ITS BACKGROUND

The education sector plays an important role in shaping students' knowledge, skills, and overall development. To fulfill this responsibility, schools must efficiently manage essential processes such as student enrollment and academic record keeping. These processes demand accuracy, speed, and organization to ensure that students, parents, and school personnel receive reliable and timely services. In many Philippine public high schools, however, these processes remain largely manual, leading to inefficiencies, delays, and errors (Dela Cruz et al., 2022).

Palale High School, a public secondary school serving its local community, exemplifies these challenges. A structured interview conducted at the school on March 14, 2026, revealed several critical operational gaps. The school has a designated registrar who manages student records and enrollment processes. Enrollment is conducted entirely face-to-face, requiring students and parents to submit physical documents such as birth certificates, report cards (SF9), and for transferees, SF10 and Good Moral Certificates. This manual setup results in long waiting times and repetitive data entry. A study by Mercado and Reyes (2023) found that manual enrollment processes in Philippine public high schools led to an average waiting time of 45 minutes per student and data encoding errors in approximately 12% of records.

The teacher evaluation process is similarly manual. Students evaluate their teachers using paper forms based on classroom management, professionalism, and instructional quality, with anonymity maintained only by omitting names. The school head manually tabulates the results, a process that takes one to two days. This

method increases workload, delays feedback, and is prone to computational errors (Fernandez & Lopez, 2024).

Furthermore, the school lacks an integrated system for tracking document requests. Commonly requested documents such as Certificates of Enrollment, SF9, SF10, and Good Moral Certificates require one to three working days to process, but students cannot monitor the status of their requests online. This leads to repeated follow-ups and confusion (Santos, 2023).

The school also has no digital repository for learning materials. It offers only the HUMSS strand in Senior High School, with a sufficient number of classrooms to accommodate its student population. Teachers rely heavily on printed modules and face-to-face distribution of learning materials. For all grade levels from Junior High School to Senior High School, teachers send learning materials directly to students, but there is no centralized platform where students can access subject-categorized lessons. Currently, only document files (such as DOCX and PDF) are allowed; videos and PowerPoint presentations are not permitted. This limitation makes it difficult to support blended or modular learning scenarios, especially when remote access to structured resources is needed (Department of Education [DepEd], 2021). A critical issue underlying all these problems is the lack of integration between enrollment and academic processes. Student information collected during enrollment is not automatically linked to teacher evaluation rosters, document request histories, or learning resource access. This disconnection results in duplicate data entry, inconsistent student profiles, and inefficiencies in monitoring academic progress (Villanueva & Garcia, 2022). Although the school has attempted to use shared spreadsheets and messaging applications to coordinate operations, these tools remain disconnected and do not provide a unified system.

Given these identified challenges, the development of a Web-Based Integrated Enrollment and Academic Management System for Palale High School is necessary. The proposed system will integrate enrollment, teacher evaluation, document request tracking, and learning management into a single web-based platform. Once a student is enrolled, their information will automatically be accessible across all modules, eliminating redundant encoding and ensuring consistency. The system will allow online submission of enrollment requirements, digital teacher evaluations with automatic computation of results, real-time tracking of document requests, and centralized access to learning materials categorized by grade level and subject. Through this integration, the system aims to improve efficiency, reduce administrative workload, enhance data accuracy, and provide better service to students, parents, teachers, and administrators. Furthermore, the implementation of this system aligns with Department of Education Memorandum No. 42, s. 2021, which encourages the adoption of automated enrollment and learning management solutions in schools.

The system will be developed using an Agile software development approach, enabling the research team to respond quickly to user feedback and deliver incremental improvements throughout the project lifecycle.

I. REVIEW OF RELATED LITERATURE (RRL)

Fernandez and Lopez (2024) conducted a systematic literature review examining web-based enrollment and academic management systems for K-12 institutions from 2020 to 2024. Their review synthesized findings from peer-reviewed journals, conference proceedings, and case studies across Southeast Asia. The authors found that schools adopting integrated web platforms reduced processing time by an

average of 68% compared to those still using manual or disconnected digital systems. Key success factors identified included user-friendly interfaces, real-time data synchronization between enrollment and academic records, and role-based access control for teachers, administrators, and students. The review concluded that integration between enrollment and academic modules is a critical success factor for school management systems, as fragmented systems often lead to data duplication and inconsistent reporting.

Dela Cruz, Santos, and Fernandez (2022) investigated the challenges faced by Philippine public secondary schools in managing enrollment and academic records manually. Through a mixed-methods study involving 45 public high schools, they documented that over 80% of administrators reported data integration problems as the primary cause of enrollment delays and grade reporting errors. The study highlighted how the lack of a centralized database forces schools to maintain separate, incompatible systems for enrollment data, grade records, and student profiles. The researchers emphasized that failure to connect and integrate student data at every level interrupt the proper flow and utilization of information. Their findings underscore the urgent need for integrated information systems that can unify administrative and academic functions within a single platform.

A study on digital enrollment systems for educational institutions described the optimization of enrollment processes through web-based technology (Materials and methods on digital enrollment system for educational institutions, 2021). The research outlined a four-phase development approach: planning (system requirements), design (model selection), coding (programming languages for structure and operation), and testing (error verification). The digital enrollment process was optimized for student registration and report generation, allowing the reduction of crowds, time, and human effort. Importantly, the system enabled

collaborative work at a distance between the institution and students, serving as a technological aid that reduces enrollment processing time in an equitable and transparent manner while guaranteeing equal access to remote education.

Mappalotteng, Fathahillah, and Punggawa (2024) developed and tested a web-based student academic grade processing information system at UPT SMP Negeri 1 Binamu. Their research addressed the problem of conventional manual documentation that made grade processing less effective and efficient, with difficult access to previous semester grades, large document storage requirements, and high risk of document loss. The system was built using MySQL, Express, React, and Node.js technology stacks with REST API architecture. Testing employed the ISO/IEC 25010 standard, yielding a functional suitability value of 1, performance efficiency with page loading under 10 seconds, very good usability, 100% reliability, and operating well for compatibility and portability. The overall test results deemed the system highly feasible for academic grade management.

Cierva and Malangsa (2023) analyzed the effectiveness of a digital enrollment and grading system implemented at Bato Institute of Science and Technology. Their comprehensive platform managed all aspects of student enrollment and grading, providing an efficient way to store and manage student data, grade reports, and enrollment records. The completely computerized process reduced the chance of human error while maintaining accurate data. The system included a backup feature for retrieving important information in case of malfunctions, ensuring data security. Two access modes were implemented: administrator (for creating user accounts, scheduling updates, system maintenance) and user (allowing staff and students to access grades and reports). Based on descriptive developmental research gathering feedback from both staff and students, the researchers concluded that the system was highly effective with a

"very good" rating, though continuous evaluation and improvement were recommended.

Anwar (2024) designed and constructed a web and mobile-based academic information system for State Elementary School 3 Pelem Pringkuku Pacitan. The existing system used Microsoft Excel for data processing, which teachers found difficult to operate, and was not integrated into an efficient and effective system, resulting in slow data retrieval and delayed teacher performance. The research aimed to help record master data (student data, principal data, teacher data, parent data, curriculum data) as well as academic data (grade data, attendance data, subject data). Using DBMS with MySQL as the data storage medium, the resulting academic information system allowed parents to monitor their children's schooling through the application. Grade and attendance management features also facilitated faster information delivery on student grades and attendance.

Villanueva and Garcia (2022) conducted a study on the adoption of school management information systems in the CALABARZON region of the Philippines. Their research identified barriers and facilitators to implementing integrated school management platforms. They concluded that "integration of enrollment and grade management modules is the single most important factor for operational efficiency." The study also found that schools lacking integrated systems spent an average of 15 additional hours per week on duplicate data entry and cross-verification tasks. The researchers advocated for school management information systems that offer a centralized platform for student and teacher data management, attendance and grade tracking, classroom management, and report generation.

Gbadamosi and Ayodele (2025) developed a University Management System (UMS) to automate key academic and administrative functions. The system provided a centralized platform to manage student records, course registration, grading, and transcript processing, reducing

reliance on manual procedures. The research addressed challenges faced by many higher institutions related to manual data handling, fragmented systems, and delayed administrative operations. Built using Python with Django framework for backend development and HTML, CSS, and JavaScript for the frontend interface, core modules included student and staff management, course enrollment, result computation, and transcript generation. Evaluation through task-based scenarios and load testing demonstrated that the UMS effectively handled multiple concurrent users, maintained data accuracy, ensured fast response times, and improved administrative task speed and reliability.

Paculan, Lacdao, De Guzman, Bautista, Paano, and Lumabi (2025) developed a web-based Guidance Record Management System (GRMS) designed to simplify record-keeping in guidance offices. The system allowed students to submit and manage their records, which counselors could access for sessions. The system also included exploratory data analysis (EDA) features for data-driven decision-making, with an admin dashboard presenting visualizations and insights using EDA with Python and a Decision Tree algorithm. Built using Laravel PHP framework, CSS and Tailwindcss for styling, JavaScript, Python's Pandas library with Matplotlib and Seaborn for visualization, MySQL database, and Apache server, the system was evaluated according to ISO 25010 standard for functionality, reliability, usability, and performance. Evaluation results revealed that both male and female users, as well as technical respondents, strongly agreed on the acceptability and usability of the system, with average mean ratings of 3.4 to 3.5, indicating consistent positive reception.

Enad and Balicoco (2023) proposed the implementation of a Management Information System for public secondary schools in the Sagbayan District of the Philippines. Their study emphasized that the absence of a computer system, combined with the complexity of school transactions, often leads to personnel being overloaded with

paperwork in storing and keeping student records. The researchers argued that school management information systems should provide a comprehensive solution that includes student and teacher data management, attendance and grade tracking, classroom management, and report generation. The proposed system aimed to modernize school administration tasks and serve as the motivating factor for designing and developing an integrated School Management Information System that could lift the transaction process of schools to a more efficient level.

Serrano and colleagues (2022, published 2023) developed and tested a web-based enrollment system for a public junior high school in Cavite province, one of the most populous public junior high schools in the area. The study was motivated by the Philippine Education System's continued reliance on manual processes during enrollment procedures, which became particularly problematic during the pandemic. While private schools managed to cope, public schools faced major problems with the enrollment process and the start of the academic year. The researchers utilized computerization principles to interrelate different but interdependent transactions, resulting in a system with well-integrated processes that performed much faster and more accurately than the manual system. Results showed that the requirements set by users were met, and the overall evaluation demonstrated that the system was effective and useful in addressing the school's enrollment needs.

Matos Jr. (2022) designed and implemented an online enrolment system for a higher education institution in Zamboanga Peninsula amidst the COVID-19 pandemic. The study identified constraints in the existing enrollment system, particularly in detecting conflicts in course schedules, availability of slots for courses offered, handling large volumes of data, and cases requiring modifications or error fixes. The methodology employed prototyping to take advantage of users' limited experience with computerized systems. The system was

implemented using Hypertext Pre-processor (PHP) programming language and MySQL database along with JavaScript, Cascading Style Sheets (CSS), JQuery, and Macromedia Dreamweaver as the integrated development environment. The functionalities of the system received approval from school administrators, allowing the program to be used in an actual enrollment setting. This study demonstrated that web-based enrollment systems can be successfully implemented even in institutions with limited prior exposure to computerized systems.

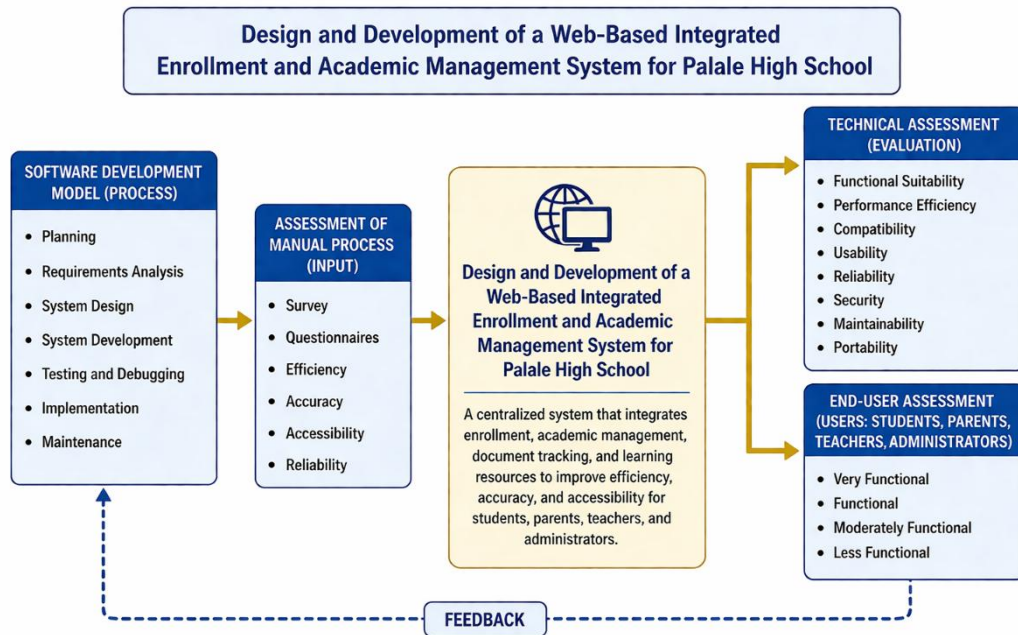
Mercado and Reyes (2023) conducted a quasi-experimental study examining efficiency gains from computerizing enrollment processes in Cavite public high schools. The research measured the impact of a web-based enrollment system on waiting times, data accuracy, and user satisfaction. Their findings revealed that manual enrollment processes resulted in an average waiting time of 45 minutes per student and data encoding errors in approximately 12% of records. After implementation of the computerized system, waiting times decreased by 68% on average, and data encoding errors were virtually eliminated. The study provided empirical evidence that computerization reduces human errors and processing time, thereby boosting productivity and resulting in high-quality output. The researchers concluded that the computerization of enrollment processes should be prioritized by public schools seeking to improve operational efficiency.

Santos (2023) conducted a needs analysis for an integrated enrollment and grade management system in rural high schools, specifically focusing on Batangas province. Through surveys and interviews with school personnel, the study revealed that 78% of enrollment staff reported difficulty retrieving student academic records due to disconnected systems, and 64% of teachers experienced delays in submitting grades because of incompatible or inaccessible record-keeping tools. The study documented that the absence of a computer system, combined with the complexity of school transactions,

often led to personnel being overloaded with paperwork in storing and keeping student records. These findings served as the motivating factor for designing and developing a School Management Information System. The study emphasized that failure to integrate enrollment and academic records leads to redundant data entry, inconsistent student profiles, and inefficiencies in managing academic progress.

Cierva and Malangsa (2023) conducted an effectiveness analysis of a digital enrollment and grading system implemented at Bato Institute of Science and Technology. Using Descriptive Developmental Research methodology, the researchers gathered feedback from both staff and students to evaluate the system's performance. The Online Enrollment and Grading System served as a comprehensive platform that helped manage all aspects of student enrollment and grading, providing efficient storage and management of student data, grade reports, and enrollment records. The system ensured that all relevant information was tracked from the beginning of a student's course until its conclusion. The completely computerized process reduced the chance of human error and maintained accurate data, and included a backup system for retrieval of important information in case of system malfunctions. Based on the results, the researcher concluded that the Online Enrollment and Grading System was highly effective with a "very good" rating, though continuous evaluation and improvement were recommended to maintain high standards of service and efficiency.

Conceptual Framework



The diagram illustrates the conceptual framework of the study titled *Design and Development of a Web-Based Integrated Enrollment and Academic Management System for Palale High School*. It presents the flow of processes involved in developing and evaluating the proposed system.

The process begins with the Agile Software Development Model which serves as the foundation of the system creation. This model emphasizes iterative development, continuous user feedback, and collaboration. The stages—planning, requirements analysis, system design, system development, testing and debugging, implementation, and maintenance—are repeated in short cycles (sprints), allowing the system to be refined based on ongoing evaluation. This ensures that the system is carefully planned, properly built, and continuously improved to meet the evolving needs of the school. Next, the Assessment of Manual Process (Input) is conducted to identify existing problems and inefficiencies in the current system. Data is

gathered through surveys and questionnaires, focusing on key factors such as efficiency, accuracy, accessibility, and reliability. This phase helps the researchers understand the limitations of the manual processes currently used in enrollment and academic management.

The gathered data then contributes to the development of the proposed system, placed at the center of the framework. The system integrates enrollment and academic management functions into a single web-based platform designed to improve overall school operations.

After development, the system undergoes Technical Assessment (Evaluation) using standard software quality criteria, including functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. This ensures that the system meets technical standards and performs effectively.

Simultaneously, an End-User Assessment is conducted among students, teachers, and administrators. The system is evaluated based on usability and functionality using rating scales such as very functional, functional, moderately functional, and less functional. This step determines how well the system meets the needs of its intended users.

Finally, a feedback mechanism is incorporated, connecting all stages of the process. Feedback from both technical evaluation and end-users is used to further improve and refine the system, ensuring continuous development and enhancement.

Statement of the Problem

The purpose of this study is to develop a Web-Based Integrated Enrollment and Academic Management System for Palale High School that will address the operational challenges currently experienced by the

school. The enrollment and academic management processes in public secondary schools require efficient coordination of student data, enrollment records, teacher evaluations, document requests, and learning resources. However, based on the current operations of Palale High School as revealed in the interview conducted on March 14, 2026, several problems have been identified that affect the overall efficiency and reliability of its services.

Specifically, this study seeks to answer the following questions:

1.) How may the Web-Based Integrated Enrollment and Academic Management System for Palale High School be designed and developed using the Agile software development approach, following the iterative stages of:

1.1. Planning

1.2. Requirements Analysis

1.3. System Design

1.4. System Development

1.5. Testing and Debugging

1.6. Implementation

1.7. Maintenance

2.) How may IT experts evaluate the technical quality of the system based on the ISO 25010 software quality model in terms of:

2.1. Functional Suitability

2.2. Performance Efficiency

2.3. Compatibility

2.4. Usability

2.5. Reliability

2.6. Security

2.7. Maintainability

2.8. Portability

3.) How may end-users, including students, teachers, and school administrators of Palale High School, assess the system based on:

3.1. Very Functional

3.2. Functional

3.3. Moderately Functional

3.4. Less Functional

4.) How does Palale High School currently manage its enrollment and academic records without an integrated system, particularly in terms of:

4.1. Processing new student enrollment and document verification

4.2. Conducting teacher evaluations and tabulating results

4.3. Tracking and fulfilling student document requests (e.g., certificates, SF9, SF10)

4.4. Storing, accessing, and distributing learning materials to students

Objectives of the Study

General Objective

The main objective of the Web-Based Integrated Enrollment and Academic Management System for Palale High School is to design and develop a centralized web-based system that will improve the management of student enrollment and academic records. This study aims to create a platform that will help to modernize enrollment

transactions, manage teacher evaluations, track document requests, and organize learning resources to make school operations faster, more accurate, and more efficient.

Specific Objectives

- To develop a system that can manage and organize student enrollment data, including personal information, required documents (birth certificate, SF9, SF10), and student type (new, transferee, regular).
- To provide an online enrollment module that allows students and parents to submit requirements digitally, reducing face-to-face transactions and long waiting times.
- To design a teacher evaluation module that enables students to submit anonymous feedback based on classroom management, professionalism, and instructional quality, with automatic tabulation of results for the school head.
- To automate the generation of document requests (Certificates of Enrollment, SF9, SF10, Good Moral Certificate) and track their processing status to reduce manual follow-ups and processing time.
- To create a learning management module that can store subject-categorized learning materials (PDF modules, PowerPoint presentations) organized by grade level and subject.
- To allow teachers to upload and manage learning resources, and students to access or download materials.
- To generate reports such as enrollment summaries, teacher evaluation results and document request logs.
- To implement secure user access with different roles such as school head, administrator, teacher, and student.

- To develop a user-friendly and efficient web-based interface that will improve the overall enrollment and academic management workflow of Palale High School.

Scope and Delimitation

This study focuses on the design, development, and evaluation of a Web-Based Integrated Enrollment and Academic Management System for Palale High School. The system is intended to address the operational gaps identified in the school's current manual processes, particularly in enrollment, teacher evaluation, document request tracking, and access to learning materials. In addition to these core functions, the system includes a Grading Module as an extra feature to provide complete academic management capability, even though grade management was not identified as a primary problem in the initial needs assessment.

Scope

The study covers the following nine (9) functional modules of the proposed system:

Authentication and Login Module - Provides role-based access for students, teachers, registrar staff, and the school head, with secure session management and password encryption.

Announcement Module - Allows administrators to post school-wide news and updates. Announcements support text and image attachments (JPEG, PNG) and include a "pinned" feature for urgent alerts.

Faculty Performance Evaluation Module - Enables students to submit anonymous ratings of their teachers based on instructional quality, classroom management, and professionalism. The system automatically tabulates results and generates summary reports for the school head.

Registrar Module - Manages student records, document request tracking (Certificate of Enrollment, SF9, SF10, Good Moral Certificate), and class/section organization (creation of sections, assignment of homeroom advisers, and maintenance of master lists).

Admission Module - Provides an online application portal for new students (incoming Grade 7 and transferees), allowing them to submit personal information and pre-enrollment documents (PSA birth certificate, Good Moral Certificate, previous report cards) digitally.

Enrollment Module - Supports online enrollment for all student types. Continuing students are automatically evaluated based on grades from the Grading Module; newcomers, returnees, and transferees undergo registrar verification of submitted SF9 and other requirements.

Learning Management Module - Serves as a centralized digital repository for learning materials organized by grade level (7-12) and subject. Teachers can upload modules and files in DOCX and PDF formats; students can view, download, or bulk-download materials for offline use.

Grading Module - Implements a term-based grade entry system (1st, 2nd, and 3rd terms). Teachers encode final grades per subject, and students can view their term-by-term academic performance through their portals.

Report Module - Generates enrollment summaries, academic performance reports (honors lists, class averages), document request logs, and consolidated faculty evaluation results. Reports can be exported for submission to school administrators and the Division Office.

Delimitation

The study does not cover the following:

Online payment processing - The system does not include any financial transaction module because public schools do not collect tuition or miscellaneous fees through the platform.

Real-time messaging or chat - No built-in instant messaging or real-time notification system (e.g., SMS, push notifications) is provided; communication remains through traditional channels (email, messaging apps, or printed notices).

Mobile application version - The system is designed as a web-based platform accessible via browsers. A dedicated mobile app for iOS or Android is not developed, though the interface is expected to be responsive for basic smartphone use.

Integration with external systems (LIS, DepEd databases) - The system operates as a standalone prototype. It does not automatically synchronize with the Learner Information System (LIS) or other DepEd-managed databases. Data entry for student records is performed manually by the registrar based on official school forms.

Parent accounts - Parents do not have dedicated login credentials. All student-related actions (enrollment, document requests, viewing grades) are performed by the student themselves, although a parent may assist offline.

Video or multimedia uploads for learning materials - The Learning Management Module accepts only DOCX and PDF files. Videos, PowerPoint presentations, and other multimedia formats are not supported, as per the existing school policy cited in the introduction.

Real-time class scheduling or attendance tracking - The system does not manage daily class schedules, teacher assignments per hour, or student attendance records. It only handles term-based grade entry and section/advisory assignments.

Automated document generation (e.g., printing SF9/SF10) - The system only tracks request statuses; the actual printing, signing, and release of physical documents are still performed manually by the registrar. The system does not auto-generate or pre-fill official DepEd forms.

Significance of the Study

This section presents the beneficiaries of the study and the benefits they will gain from the development of the capstone project titled Design and Development of a Web-Based Integrated Enrollment and Academic Management System for Palale High School.

Researchers: The system will help the researchers gain practical knowledge and experience in developing a real-world web-based system. It enhances their skills in system design, database management, and problem-solving. This study also allows them to understand the challenges in managing student enrollment and academic records in a public high school setting and how technology can be used to improve school operations.

Palale High School: The system will benefit the school by improving the overall workflow of enrollment and academic record

management. It speeds up enrollment transactions and document request processing. With a centralized and integrated system, the school can easily monitor student data, manage teacher evaluation results, track document requests, and organize learning resources, resulting in more reliable and efficient service for students.

Students: The system will help students by providing an easier and faster way to enroll, request academic documents, and access learning materials online. It reduces the need for repeated face-to-face visits to the school. Students can also submit teacher evaluations anonymously, ensuring their feedback is heard while saving time. The online document request tracker allows them to know when their certificates and forms are ready for pickup.

Teachers: The system will assist teachers in managing their evaluation results more efficiently. Instead of waiting for manual tabulation that takes one to two days, teachers can view their evaluation reports instantly through the system. Teachers can also upload learning materials (modules in DOCX and PDF format) to the learning management module, making it easier for students to access resources anytime, whether for face-to-face or modular learning.

School Head and Administrators: The system will help school administrators by providing a centralized dashboard for monitoring enrollment data, teacher evaluation summaries, document request logs, and learning resource usage. It simplifies the validation of enrollment documents, reduces manual paperwork, and generates accurate reports that can be submitted to the Division Office. This leads to better decision-making and improved school management.

Future Researchers: This study will serve as a useful reference for future researchers who plan to develop similar systems related to enrollment management, academic record keeping, or school management information systems. It provides insights into system integration, automation, and web-based solutions, which can be used as a guide for

further improvements and innovations in the field of educational technology.

Definition of Terms

For the purpose of this study, the following terms are defined operationally:

Academic Management- Refers to the processes related to maintaining and handling student academic records, including teacher evaluations, document requests, grading, and learning resources. In this study, academic management is integrated with enrollment functions within a single web-based system.

Certificate of Enrollment- A document issued by the school to certify that a student is officially enrolled for a particular grade level and school year. The proposed system allows students to request this certificate online and track its processing status.

Document Request Tracker- A module within the proposed system that allows students to request academic documents (e.g., Certificate of Enrollment, SF9, SF10, Good Moral Certificate) online and monitor the status of their requests (e.g., pending, processing, ready for pickup).

Enrollment- The process of registering students for a particular grade level and school year. In this study, enrollment includes early registration (for Grade 7 and Grade 11) and regular enrollment during Brigada Eskwela. The proposed system provides an online enrollment module where students and parents can submit requirements digitally.

Good Moral Certificate- An official document issued by the school certifying that a student has demonstrated good character and conduct. This document is commonly requested by transferees and graduates. The proposed system allows online request and tracking of this certificate.

Grading Module- An additional feature of the proposed system that allows teachers to encode term-based grades (1st, 2nd, 3rd term) and enables students to view their academic performance per subject. This module was included to provide complete academic management capability, even though grade management was not identified as a primary problem in the initial needs assessment.

ICT Coordinator- The school personnel responsible for managing information and communication technology resources at Palale High School. In the current manual setup, the ICT Coordinator helps manage student records and enrollment data. In the proposed system, the ICT Coordinator serves as an administrator.

Integrated System- A platform that combines multiple functions into a single, unified application. In this study, the integrated system combines enrollment, admission, teacher evaluation, document request tracking, grading, learning management, announcements, and reporting into one web-based platform, allowing data to be shared across modules without redundant entry.

Learning Management Module- A component of the proposed system that serves as a digital repository for learning materials. Teachers can upload subject-categorized resources in ****DOCX and PDF formats only****. Students can view, download, or bulk-download these materials for online or offline use. The module does not support video, audio, or PowerPoint file uploads, consistent with the existing school policy.

Manual Process- Refers to the current method of performing school transactions using paper forms, physical signatures, face-to-face submissions, and manual tabulation. Examples include manual enrollment, paper-based teacher evaluation, untracked document requests, and printed module distribution.

SF9- The official school form known as the "Report Card." It contains a student's grades, attendance, and behavior for a given grading period. Students may request a copy of their SF9 through the proposed document request module.

SF10-The official school form known as the "Permanent Record" or "Student Permanent Academic Record." It contains a student's complete academic history from elementary to secondary school. Transferees are required to submit SF10 during enrollment.

School Head- The principal or highest-ranking administrator of Palale High School. In the current system, the school head manually tabulates teacher evaluation results. In the proposed system, the school head can view automatically generated evaluation reports and monitor system-wide data.

Teacher Evaluation- The process of assessing teacher performance based on criteria such as classroom management, professionalism, and instructional quality. Students currently submit paper-based evaluations; the proposed system automates anonymous submission and instant tabulation.

Web-Based System- A software application accessed through a web browser over the internet or a local network. The proposed system is

web-based, allowing students, parents, teachers, and administrators to access enrollment and academic management functions anytime and anywhere.

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CHAPTER 2

METHODOLOGY

This chapter presents the systematic procedures and methods used in conducting the study. It includes the research design, research locale, respondents of the study, research instruments, data gathering procedures, data analysis, and ethical considerations.

Research Design

This study utilized the Descriptive Developmental Method, which combines both descriptive and developmental approaches. The descriptive aspect focused on gathering relevant data through interviews and surveys to understand the current processes, challenges, and needs of Palale High School in managing student enrollment, teacher evaluations, document requests, and learning

resources. The developmental aspect involved the design, development, and testing of the proposed system.

The descriptive method was used to evaluate the existing manual processes and identify common problems such as long waiting times during enrollment, manual tabulation of teacher evaluation results, lack of a document request tracking system, and absence of a centralized digital repository for learning materials. Data were collected from school administrators, registrar staff, teachers, and students to ensure that the system requirements were based on actual user needs.

On the other hand, the developmental method was applied in creating the proposed system. The study adopted the Agile Software Development Model which emphasizes iterative development, continuous feedback, and collaboration with users. This approach allowed the researchers to refine the system through multiple stages (planning, requirements analysis, system design, system development, testing and debugging, implementation, and maintenance), ensuring that improvements were made based on user evaluation and testing.

By combining these two methods, the researchers ensured that the system was both relevant to user needs and effectively developed.

Research Locale

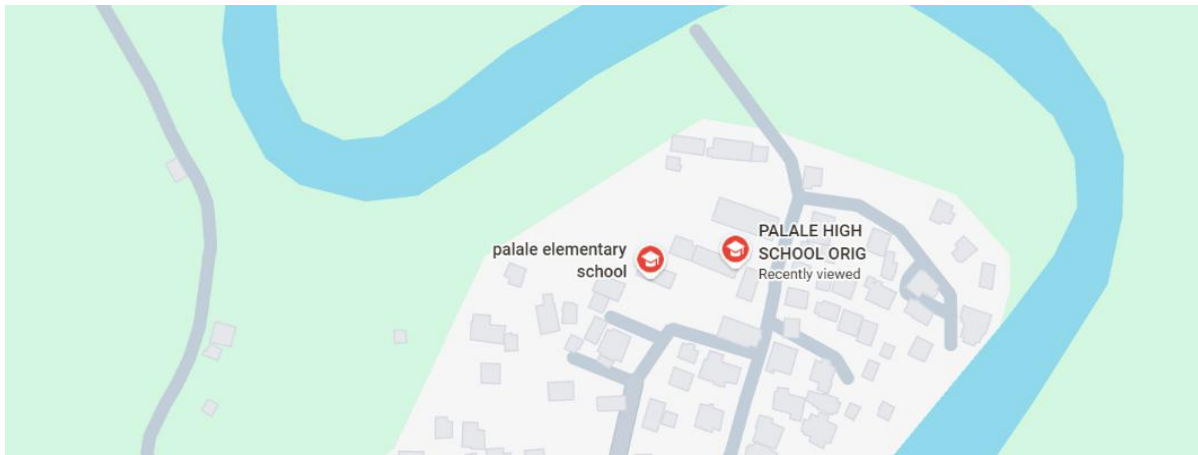


Figure 2 Location Map

This study was conducted at Palale High School, located in Barangay Palale, General Tinio, Nueva Ecija, Philippines. The researchers selected this location not only for its accessibility but also for its direct relevance to the objectives of the study. The school currently relies on manual processes for enrollment, teacher evaluation, document requests, and distribution of learning materials, making it an ideal setting for the development and testing of a web-based integrated enrollment and academic management system.

Respondents of the Study

The respondents of this study were selected based on their direct involvement in the school's enrollment and academic management processes. They included administrators, registrar staff, teachers, and students who actively participate in the school's daily academic and administrative operations.

The School Head is responsible for overseeing the system and making high-level decisions. Registrar staff assist in processing academic documents and validating enrollment requirements. Teachers are assigned based on subject specialization (e.g., Mathematics, Science, English, Filipino). Each teacher handles a specific subject across multiple grade-level sections, managing class records, uploading learning materials, and monitoring student academic progress. Students use the system for enrollment, accessing learning materials, and requesting academic documents.

A total of 480 respondents participated in the study, consisting of 3 administrators, 2 registrar staff, 15 teachers, and 460 students (345 from Junior High School and 115 from Senior High School). This distribution ensures a realistic representation of the school's organizational structure, with approximately three-fourths of the student population coming from JHS, which mirrors the typical enrollment pattern in many public high schools. The student-to-teacher ratio is approximately 31:1, which is well within the realistic range for public secondary schools and allows teachers to manage class records and academic monitoring effectively. The number of teachers (15) is sufficient to cover subject specializations across both JHS (eight subject areas) and SHS (core and specialized subjects) without overloading individual teachers. The sample size strengthens the validity of the findings and supports an accurate assessment of the system's functionality, usability, and effectiveness.

Research Instrument

The primary research instrument used in this study was the structured interview guide and survey questionnaire. The researchers conducted interviews with selected respondents from Palale High School. The interview guide consisted of carefully prepared questions

designed to gather data relevant to the development of the proposed Web-Based Integrated Enrollment and Academic Management System.

The interview guide and survey questionnaire focused on six key areas. First, the current manual processes for enrollment were examined, including document submission, verification procedures, and waiting times. Second, the manual teacher evaluation process was assessed, particularly the use of paper forms and the time required for tabulation. Third, the existing process for document requests was explored, covering certificates, SF9, SF10, and Good Moral Certificates. Fourth, the school's method of storing and distributing learning materials was investigated. Fifth, the challenges and limitations encountered by school personnel, teachers, and students were identified. Finally, respondents were asked about the features and improvements they expect from the proposed computerized system.

The questionnaire included both close-ended and open-ended questions to allow respondents to share detailed insights, experiences, and suggestions. For the technical evaluation, a separate questionnaire based on the ISO 25010 software quality model was administered to IT experts who were consulted separately for system evaluation. For end-user assessment, a functional rating scale (Very Functional, Functional, Moderately Functional, Less Functional) was used.

To ensure the reliability and relevance of the instrument, the interview guide and questionnaires were reviewed and validated by the research adviser. The data collected played a significant role in guiding the system's development and ensuring that it effectively addresses the actual needs and operational requirements of the school.

Data Gathering Procedure

The initial phase of this research involved seeking approval from the research adviser to proceed with the data collection process. Upon receiving approval, the researchers designed a structured questionnaire using Google Forms to efficiently gather data from selected respondents of Palale High School. Prior to distributing the questionnaire, formal permission was secured from the school administration to conduct the study.

The questionnaire link was then distributed to the respondents through appropriate channels (e.g., email, messaging apps). Before answering the survey, the respondents were provided with a brief introduction explaining the purpose of the study and were asked to give their informed consent.

Respondents were given sufficient time to complete the questionnaire at their convenience. The instrument included both close-ended and open-ended questions to allow participants to provide detailed responses regarding the current manual processes (enrollment, teacher evaluation, document requests, learning resource access), their challenges, and their suggestions for improvement.

Throughout the data gathering process, the researchers ensured that all responses were properly recorded and automatically stored through the platform. The collected data were carefully reviewed and organized for analysis. Confidentiality and privacy of the respondents were strictly maintained, and all information gathered was used solely for academic and research purposes.

Data Analysis

In this study, the researchers used descriptive data analysis to evaluate and summarize the information gathered from the respondents. The primary goal was to understand the current operational challenges faced by Palale High School in managing enrollment, teacher evaluations, document requests, and learning resources.

After organizing the responses from the data collected, common issues such as the time spent on manual enrollment (long queues, repetitive data entry), the 1-2 day manual tabulation of teacher evaluation results, the lack of a document request tracking system (students cannot monitor status), and the absence of a centralized digital repository for learning materials were carefully examined. To better understand the trends and patterns, the data was also presented using tables and charts to make it easier to interpret.

For the technical evaluation, data from IT experts using the ISO 25010 criteria were analyzed by computing mean scores for each quality characteristic (functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, portability). For the end-user assessment, responses from students, teachers, and administrators were analyzed using the four-point functional scale (Very Functional, Functional, Moderately Functional, Less Functional).

This analysis guided the formulation of the system's requirements and features, ensuring that the proposed Web-Based Integrated Enrollment and Academic Management System effectively address the identified problems and enhances the school's administrative and academic processes.

Ethical Consideration

This study strictly adhered to ethical standards to protect the rights, privacy, and dignity of all participants involved in the design, development, and evaluation of the Web-Based Integrated Enrollment and Academic Management System for Palale High School. The researchers recognized that the study involved human participants, including students, teachers, administrators, registrar staff, and IT experts, and therefore upheld the highest ethical principles throughout the entire research process—from data collection to system deployment and evaluation.

The following ethical protocols were observed:

Informed Consent: All participants were informed of the purpose of the study, the voluntary nature of their participation, and their right to withdraw at any time without penalty.

Confidentiality and Anonymity: Personal information and responses were kept confidential. Anonymity was maintained, especially for student evaluation responses.

Respect for Participants: Researchers treated all participants with respect and courtesy.

No Harm: The study did not cause any physical, psychological, or social harm to participants.

Data Security: All collected data were stored securely and accessed only by the research team.

Academic Integrity: The researchers properly cited all sources and avoided plagiarism.

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